

Логарифмически вогнутые меры

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The uniform measure on a convex body and the Gaussian measure are two instances of one class. Inside that class one has concentration, exponential tails and the whole “function versus body” dictionary, so statements about convex bodies and about random variables become one and the same statement.

Definition 1. A measure μ on $\mathbb{R}^n$ is **log-concave** if $\mu(\lambda A + (1 - \lambda)B) \geq \mu(A)^\lambda \mu(B)^{1-\lambda}$ for all compact A, B and $\lambda \in (0, 1)$.

Definition 2. A log-concave probability measure μ is **isotropic** if its barycentre is 0 and $\int \langle x, \theta \rangle^2 d\mu = L_\mu^2$ for every unit vector θ .

Theorem 1 (Borell 1974). A measure on $\mathbb{R}^n$ which is not supported on a hyperplane is log-concave if and only if it has a log-concave density.

Theorem 2 (Borell’s lemma). Let μ be a log-concave probability measure, A a symmetric convex set with $\mu(A) = \theta > \frac{1}{2}$. Then for $t \geq 1$

$$\mu(\mathbb{R}^n \setminus tA) \leq \theta \left(\frac{1 - \theta}{\theta} \right)^{(t+1)/2}.$$

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1. Prove the easy half of Theorem 1: a measure with a log-concave density is log-concave. (*Hint: this is Prékopa–Leindler with $f = \mathbf{1}_A$, $g = \mathbf{1}_B$.*) Give an example of a log-concave measure supported on a hyperplane, and explain why it is excluded.
 2. (Borell 1974) Prove Theorem 2. (*Hint: check the inclusion $A \supseteq \frac{2}{t+1}(\mathbb{R}^n \setminus tA) + \frac{t-1}{t+1}A$ and apply log-concavity.*)
 3. Deduce from Theorem 2 that a log-concave probability measure has exponential tails: there is $c > 0$ with $\int e^{c\|x\|} d\mu(x) < \infty$. Deduce that all moments are finite.

4. Prove that linear images and marginals of log-concave measures are log-concave, and that the uniform measure on a convex body is log-concave. Deduce that every one-dimensional marginal of the uniform measure on a convex body has a log-concave density.
5. Let f be a log-concave probability density on $\mathbb{R}$ with mean m . Prove that f is unimodal and that $\mathbb{P}(X \geq m) \geq 1/e$. (*Hint: this is one-dimensional Grünbaum; compare with the exponential distribution.*)
6. (warm-up) Compute the density of the sum of two independent random variables uniform on $[0, 1]$, check directly that it is log-concave, and explain which general theorem this illustrates.